

Assignment 1

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Introduction

This project involves the design and development of a website for S.G.V. Tegel, a close-knit group of friends from Enschede, actively engaged in the study association Inter-Actief and broader university activities. With a recent move toward a more structured organization, including a board, meetings and regular events. Tegel requires a centralized platform to streamline the planning and management of their activities. The primary objectives of the website are to display and manage events, allow users to sign up while providing organizers with relevant registration details, and share announcements and newsletters. The platform will serve as a vital tool to help the group remain organized and connected.

Goals

The primary goals of the project are to enable the display and management of activities organized by S.G.V. Tegel, facilitate user sign-up for these activities while providing organizers with relevant registration information, and allow users to check general announcements and newsletters to stay informed about updates and upcoming events.

Functional Requirements

These requirements define the core features and functionalities that the system must provide to meet user and administrative needs. They include essential tasks such as user registration, account management, activity enrolment, and the ability to view announcements and newsletters. These functionalities directly support the primary goals of the platform.

Related to Management of Users

These requirements focus on user account creation, modification, and administration.

Requirement	User Story	MoSCoW
The system shall allow users to register new accounts	As a USER, I want to register an account so that I can sign up for activities.	Must
The system shall support email verification for new accounts	As a USER, I want to activate my account via a link in the confirmation email.	Must
The system shall notify users of duplicate email registrations	As a USER, I want to be notified if my email address is already registered, so that I know I have an account	Should
The system shall allow users to modify their account information	As a USER, I want to change my account information.	Must
The system shall allow administrators to manage user accounts	As an ADMINISTRATOR, I want to manage user accounts so that only authorized persons can access the system and participate.	Must

Related to Management of Activities

These describe how activities are created, displayed, and managed within the system.

Requirement	User Story	MoSCoW
The system shall provide CRUD operations for activities	As an ADMINISTRATOR, I want to create, edit, update and delete activities because it allows me to organise and manage upcoming events efficiently.	Must
The system shall display upcoming events to users	As a USER, I want to view a list of upcoming events or activities so that I can stay informed about what is happening in the group.	Must
The system shall show detailed activity information	As a USER, I want to view detailed information about each activity so that I can make an informed decision about attending.	Must
The system shall allow users to enroll in activities	As a USER, I want to enrol in an activity so that I can participate in events that interest me.	Must
The system shall allow users to cancel activity enrollments	As a USER, I want to be able to cancel my enrolment if my plans change so that my spot can be given to someone else.	Should

Related to Activity Registration

These define how users interact with the registration process and provide personal preferences.

Requirement	User Story	MoSCoW
The system shall allow users to specify preferences during registration	As a USER, I want to specify options like ordering food during registration because it ensures my personal preferences and needs are considered.	Should
The system shall allow	As a USER, I want to be able to leave a	

users to add notes to their registration	note when enrolling for activities so I can add additional information if necessary.	Should
The system shall display participant preferences to administrators	As an ADMINISTRATOR, I want to see participants' food preferences and notes because it helps in making proper arrangements for each event.	Must

Related to Announcements and Newsletters

These requirements cover how information is communicated from administrators to users.

Requirement	User Story	MoSCoW
The system shall allow administrators to post announcements	As an ADMINISTRATOR, I want to post announcements because it allows me to quickly inform all members about important updates and events.	Must
The system shall display announcements and newsletters to users	As a USER, I want to view announcements and newsletters because it keeps me updated on group news and upcoming activities.	Must

Non-Functional Requirements

Non-functional requirements describe how the system should operate rather than what it should do. They focus on attributes like security, performance, usability, and maintainability to ensure the system is reliable, efficient, and user-friendly. These requirements are critical for providing a high-quality, scalable and a secure user experience.

Related to Security

These requirements ensure the system protects user data and prevents unauthorized access.

Requirement	User Story	MoSCoW
The system shall restrict actions to authorized users who are logged in	As an ADMINISTRATOR, I want to restrict action to authorised and logged-in users to prevent unknown users from interfering.	Must
The system shall ensure sensitive data such as usernames and passwords are stored encrypted	As a USER, I want the system to safely store sensitive data such as passwords so that I do not lose my account.	Must
The system shall automatically log out users after a period of inactivity	As an ADMINISTRATOR I want the system to automatically log out users after a period of inactivity, so the system uses fewer resources.	Should
The system shall support email-based account authorization	As a USER, I want to verify my account by email so I can link back and recover my account.	Could
The system shall be protected against SQL-injection	As an ADMINISTRATOR I want the system to be protected against SQL-injection, so users with malicious intent cannot harm the database.	Must

Related to Performance and Availability

These define how the system should behave under load and its expected uptime and responsiveness.

Requirement	User Story	MoSCoW
The system shall handle up to 500 users	As an ADMINISTRATOR I want the system to handle up to 500 users at the same time so there is room for future growth.	Could

The system shall handle high user load during peak sign-up times	As an ADMINISTRATOR I want the system to handle a high simultaneous user load, so peak sign-up times for limited events cause no problems.	Must
The system shall be reachable 95% of the time over a monthly period	As a USER I want the system to be available 95% of the time, so I can access the website whenever I want.	Must
The system shall provide reliable service 95% of the time	As a USER I want reliable service 95% of the time, so I can effectively use the webservice whenever I want.	Must
The system shall clearly communicate with the user in case of system failure through clear error messages	As a USER I want the system to give me clear error messages in case of a failure, so I know something went wrong.	Must
The system shall clearly communicate error messages with the administrator in case of system failure	As an ADMINISTRATOR I want the system to display clear error messages in case of failure, so I know where and when something went wrong.	Must

Related to Maintainability and Extensibility

These requirements support future development and ease of system updates or expansions.

Requirement	User Story	MoSCoW
The systems design and components shall be well-documented, supporting maintenance	As a DEVELOPER I want the systems design and components to be well-structured and documented, so I can easily understand the system.	Should
The system shall be structured to allow easy future expansion of features and functionality	As a DEVELOPER I want the system to be structured for future expansion, so I can easily add new features.	Should

The system shall be designed to support future growth and capacity	As a DEVELOPER, I want the system to be structured for future growth and capacity so I can increase the server capacity if needed.	Should
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Related to Compatibility and Usability

These ensure the system works across platforms and is intuitive for both users and administrators.

Requirement	User Story	MoSCoW
The system shall be easy to use, requiring no prior experience for users	As a USER I want the system to be easy to use, so I can make full use of all provided services without prior experience.	Should
The system shall be simple to use for admins, requiring minimal training in administrative activities	As an ADMINISTRATOR I want the administrative activities to be simple to use, so I can make full use of all features with minimal training.	Should
The system shall function correctly across all major web browsers	As a USER I want to access the system across all major web-browsers, so I can choose my own favorite browser to work with.	Must

Wireframe

The mock-up maintained a consistent orange, white and black colour theme which was fitting for Tegel's aesthetic and theme.

Each page has a sticky navigation bar that contains a logo of S.G.V. Tegel and four pages, which are Home, Events, Admin and About. The top right contains two buttons for login and signs up. Each page also contains a footer which contain pictures which are button links for Tegel's social media. The address and logo are also displayed on the middle and right of the footer, respectively.

HOME PAGE: The homepage is the page that a user is directed to when they access TEGEL website. They are greeted to a slideshow, and below that sections named About, Your events, and Newsletter. This is the page that is used mainly to view announcements and newsletters.

SIGN UP/LOGIN PAGE: The user can either sign up or login based on whether they have an approved S.G.V Tegel account. The pages prompt the user to enter their (un)registered email address and password. The login page has a link to contact admin in case the user has forgotten their registered password. Both pages also contain links to each other in case the user already has an account/doesn't already have an account.

PERSONAL DETAILS PAGE: Once the user has signed up, they are forwarded to the personal details page. They are then required to enter their personal details such as name, DOB, contact email address, phone number and any dietary restrictions. This is a prerequisite.

EVENTS PAGE: This page is used to display upcoming events. The user can register for events through this page. The functionality of this page is to view event details and to register for the event.

EVENT DETAILS PAGE: The event details page contains specific details of an event. Members and administrators can also edit the event on this page.

ABOUT PAGE: The about page shows a little paragraph about S.G.V Tegel with pictures of the team.

ADMIN PAGE: This page contains tasks the admin can perform, such as viewing member list, searching for a member, deleting a member's account, view event list, create and edit events and delete event.

PERSONAL INFO PAGE: This page displays the personal information of a registered user, and the ability to edit the personal details including the profile picture. The page also contains a section to display the events the user has signed up for.

Sprint Review

Feedback Received

The system should support distinct account types for Admins, Members, and Users, with the possibility of assigning nicknames during account creation (nicknames do not need to be unique). A primary admin account must exist with full privileges to manage other admin

roles. While payment processing is not required, implementing a "Forgot Password" feature would be a valuable addition.

The interface should maintain strong visual contrast while ensuring the default design is user-friendly. Admins should have the ability to search and edit announcements, and even though Tegel events are public, users must log in to access event details. Additionally, a dedicated section should be available for Admins to upload newsletters and announcements, allowing inclusion of text, images, event details, and associated costs.

The stakeholders did like the interface design, the accessibility and overall aesthetic offered by the mock wireframe.

Sprint Retrospect

What Went Well?

- Defined clear user roles including Admin, Member, and User.
- Account creation flow was enhanced with the optional nickname field.
- Established a structure for Admins to upload newsletters and announcements with rich content (text, images, cost).
- Public events were implemented successfully with login requirements intact.
- UI design included a high-contrast mode for better accessibility.

What could have gone well?

- Forgot Password functionality was not prioritized early to give authority to only admin to change password directly, though it would improve usability.
- Admin search and edit capabilities for announcements were only partially integrated.
- Initial role management setup lacked a clear method to assign admin rights from a central account.

What do we want to try next?

- Improve the announcement search/edit interface for admins.
- Refine visual design to balance high contrast with standard aesthetics.

What puzzles us?

- How to ensure secure and scalable role-based permissions without overcomplicating the UI.
- Whether allowing nickname reuse could lead to confusion in event or registration contexts.

- How to handle newsletter uploads with large file sizes or mixed media content efficiently.